

Gating Benefits Depend on the Backbone: An Ablation Study of a Mamba-3 Block under Matched-Noise Cross-Condition Drift

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Abstract

The robustness of Mamba-style selective state-space models (SSMs) is often attributed to input-dependent selectivity. We test this attribution with a three-step ablation chain on one controlled testbed: a Mamba-3 block, bearing vibration signals (XJTU-SY, one speed-and-load transition), and noise-aware training at matched signal-to-noise ratio. Decision thresholds for each step were written down before the corresponding grid was run (internal pre-specification, not an external registry). First, replacing the input-dependent parts of Δ , A , B and C by learned constants—while the input-dependent trapezoidal weights and rotation angles of the Mamba-3 scan, the convolutional stem and the output gate are kept—costs at most about 5 pp under deep noise (paired intervals include zero), and the frozen block still outperforms a parameter-matched time-invariant S4D by 14–20 pp. Under the pre-specified criterion these four projections are therefore not the primary contributor; the remaining input-dependent scan terms were not tested. Second, removing the multiplicative SiLU gate from the frozen block costs 9–24 pp and returns it to roughly the level of S4D. Third, adding one gate branch to S4D recovers only 17–20% of the gap to the frozen block (the seed interval reaches the pre-specified 0.3 bound under last-epoch reporting); a host-by-gate contrast of 5–23 pp indicates that, in this design and training budget, the gate’s benefit depends strongly on the host. Removal and addition are not symmetric interventions, so this does not show that the gate cannot be transplanted. A separate, shorter per-class grid shows the gate-

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related differences concentrating on the minority inner-race class. Values are best-epoch macro-F1 on the evaluation condition; the central contrasts for four arms are re-assessed under last-epoch reporting. The evidence is conditional on one block, one condition pair with four bearings per side, five seeds and matched-noise training.

Keywords: state-space models, Mamba, input selectivity, gating, robustness, ablation, distribution shift, matched-noise training

1. Introduction

Selective state-space models (SSMs) of the Mamba family have rapidly become default sequence backbones, and a common narrative has formed around *why* they work: input-dependent selectivity—the ability to modulate state dynamics per token—is credited with filtering noise, adapting to distribution shift, and outperforming time-invariant predecessors such as S4D [1]. [2, 3] This belief motivates a growing family of “selective” architectures, yet it sits in unresolved tension with recent evidence: the first systematic S4D-versus-Mamba comparison on time-series classification finds the *simpler, time-invariant* model consistently superior [4], and theoretical analyses identify expressivity regimes where Mamba is provably weaker [5]. Which account is right—and, more importantly, *what exactly should be credited* when a Mamba-style block does prove robust? This paper addresses the second question for one block and one testbed, with a controlled ablation chain whose decision thresholds were fixed before each grid was run. The testbed is bearing vibration diagnosis under operating-condition drift and matched noise, in which the robustness gap is large (up to 18 pp), the evaluation labels are identical across all 865 cells (hash-verified), and the block can be dissected.

Our approach is deliberately *attributional* rather than *methodological*: we propose no new architecture and claim no state-of-the-art. Instead we take a regime in which a Mamba-style block exhibits a large, stable robustness advantage over a time-invariant SSM, and ask which ingredient earns it. The chain has three steps, each with adjudication thresholds fixed before the corresponding grid was run:

1. **Test the classical selectivity terms (necessity test).** We construct a *frozen* arm: the input-dependent parts of Δ , A , B and C

are replaced by learned constants. The Mamba-3 scan also takes an input-dependent trapezoidal weight and input-dependent rotation angles; these, the convolutional stem, the multiplicative SiLU gate, the state parameterisation, normalisation and projections are retained (Section 3.2). If these four terms drive robustness, the frozen arm should collapse toward the time-invariant baseline.

2. **Localise the necessary component.** On the frozen base, we remove one component at a time (convolutional stem; SiLU gate; heavy-tailed state parameterisation) under a single-variable discipline checked key-by-key on the state dictionary (Section 3.2), and adjudicate each arm against pre-specified Δ -thresholds.
3. **Test sufficiency by constructive grafting.** The component identified as necessary is added to plain S4D as a gate branch, and a pre-specified recovery ratio decides whether “removal collapses” is matched by “addition restores”.

The outcome is a conditional pattern. Freezing the input-dependent Δ , A , B , C projections costs at most about 5 pp in deep noise (paired intervals include zero) while the block retains a 14–20 pp advantage over S4D (best-epoch and last-epoch reporting, Section 4.4): under the pre-specified criterion these projections are not the primary contributor, although the trapezoidal and rotation terms of the Mamba-3 scan were left input-dependent. Removing the SiLU gate costs 9–24 pp and takes the block to roughly the level of plain S4D or below. Adding a gate branch to S4D recovers only $R = 0.17$ of the gap (0.20 under last-epoch reporting; the seed interval reaches the pre-specified 0.3 bound), with essentially no recovery at the deepest adjudicated noise level. Removal and addition are not mirror-image interventions (they differ in width, projection and parameter count), so we read the pair as evidence that the gate’s benefit depends strongly on the host in this design, not as proof that the gate cannot be transplanted. A per-class analysis on a separate, shorter grid indicates that the gate-related differences concentrate on the minority class.

Contributions. (i) A controlled ablation of the classical selectivity terms (Δ , A , B , C) of a Mamba-3 block under distribution drift and matched noise, in which the block’s deep-noise advantage over S4D survives freezing; the frozen arm retains other input-dependent scan terms, and BM3 versus frozen is not re-estimated under last-epoch reporting. (ii) Component-level evidence that the output gate is the one ablated component whose removal causes a

65 large, one-signed loss, that one graft design recovers little of the gap, and
66 a coarse host-by-gate interaction estimate. (iii) A reporting-rule sensitivity
67 analysis (best epoch on the evaluation condition versus final epoch) show-
68 ing that absolute levels shift by 1.4–9.5 pp while the central contrasts keep
69 their sign. (iv) A documented pre-specification and review trail—evaluation-
70 label hashes, thresholds written before each grid, and agent-assisted review—
71 including the incidents in which automated adjudication departed from the
72 pre-specified criteria; we release the trail.

73 2. Background and Related Work

74 2.1. Time-invariant and selective SSMs

75 Diagonal structured SSMs (S4D) [6] evolve a linear state with input-
76 independent parameters (A , B , C , Δ), acting as a bank of learned linear
77 filters. Mamba-style blocks [1] make Δ , B , C functions of the input token
78 (selectivity), wrap the scan in a block containing a short causal convolu-
79 tional stem, a multiplicative SiLU gate on a parallel branch, and residu-
80 al/normalisation structure; the concrete block used throughout this paper
81 follows the Mamba-3 formulation [7]. The narrative crediting selectivity for
82 robustness draws on theory showing input-dependent dynamics can represent
83 discontinuity-sensitive projections and counteract memory decay [8], and on
84 [9]. We treat these as *hypotheses to be tested*, not established attribution-
85 s: the theoretical results concern representability, which does not by itself
86 determine which component earns empirical robustness in a trained block.

87 2.2. The unresolved comparison

88 Empirically the picture is contested. A systematic 59-dataset compari-
89 son finds S4D-based models superior to Mamba variants on clean time-series
90 classification [4]; theoretical work identifies long-range regimes where Mam-
91 ba is weaker than S4D [5]. Meanwhile, application literatures—including
92 machine-condition monitoring—contain many Mamba-hybrid methods re-
93 porting noise robustness [10, 11, 12, 13, 14], but none, to our knowledge,
94 isolates *which component* of the block is responsible under controlled con-
95 ditions. Our clean-data results agree with the S4D-superiority finding (S4D
96 matches or slightly exceeds the Mamba-style block on clean data). Whether
97 the disagreement in the literature reflects a regime difference is not settled
98 by a single testbed; what we can report is that, in the drift $\times$ noise regime

studied here, the Mamba-style block’s advantage over S4D is not removed by freezing the classical selectivity terms.

2.3. Testbed rationale

We use bearing vibration diagnosis under cross-condition drift (speed and load both change) as the testbed, for three properties that matter to the attribution question rather than to the application: (a) the distribution shift is physically generated (a change of rotation speed and radial load between recorded conditions), not simulated; (b) fault signatures are sparse transients—precisely the discontinuous structures selectivity theory predicts selective SSMs should excel at [8], which makes it a plausible arena for the selectivity hypothesis; and (c) the robustness gap between the Mamba-style block and S4D is large (up to 20 pp), giving the attribution chain high signal. The testbed is two-class (details in Section 3.1); the scope of all claims is bounded accordingly (Section 6).

3. Experimental Protocol

3.1. Testbed, arms, and budget

Vibration windows from the XJTU-SY run-to-failure dataset [15] are split by operating condition: models train on one operating condition (37.5 Hz, 11 kN) and evaluate on another (40 Hz, 10 kN), so speed and load both change (cross-condition), with additive white Gaussian noise applied at matched SNR to training and evaluation ($\{\text{clean}, +10, +6, 0, -2, -6, -10 \text{ dB}\}$). The task is binary fault classification (inner-race vs. outer-race dominant classes; class imbalance makes IR the hard class, Section 4.5). All arms share an identical training budget (50 epochs, 5 seeds; the per-class grid of Section 4.5 used 20 epochs), identical splits, and identical evaluation labels, verified by a hash of the label vector: every reported cell carries `eval_sha = 6c20b367...` (Appendix A), recomputed independently during adjudication. The hash certifies the label vector, not the input windows. In total the study comprises 865 cells across five pre-specified grids (baseline grid 475; frozen-selectivity 105; component ablation 175; graft 35; per-class 75).

Composition of the data. Training uses the bearings of the 37.5 Hz/11 kN condition whose failure mode is inner race or outer race (Bearing2.1 is labelled IR; Bearing2.2, 2.4 and 2.5 are labelled OR); evaluation uses those of the 40 Hz/10 kN condition (Bearing3.3 and 3.4 are IR; Bearing3.1 and 3.5 are OR). Bearings with cage or mixed failure modes are excluded. Windows

are non-overlapping segments of 2048 samples of the horizontal channel taken from the fault phase, which is delimited by a kurtosis/RMS onset criterion with a last-15% fallback; the evaluation set holds 3,504 OR and 1,744 IR windows, the training set 4,624 OR and 624 IR windows. Two consequences matter for interpretation. The IR class is learned from a single training bearing, so class-level results may reflect bearing-specific generalisation. And the five seeds resample initialisation, batch order and noise but not bearings: there are four independent bearings on each side, and none of the intervals reported here captures bearing-level uncertainty.

The principal arms are summarised in Table 1.

Table 1: Principal experimental arms and parameter counts.

Arm	Description	Params
BM3	Mamba-style block stack (conv stem $k=7 \rightarrow$ selective scan $\rightarrow$ SiLU gate $\rightarrow$ projection), with and without an auxiliary physics loss (Section 3.4)	177,938
BM3-frozen	BM3 with the input-dependent parts of Δ , A , B , C replaced by learned constants; trapezoidal weight, rotation angles, stem, gate and the remaining components unchanged (Section 3.2)	112,410
S4D	Parameter-matched diagonal time-invariant SSM (LayerNorm; no BatchNorm anywhere in either SSM arm)	100,162
S4D-wide	S4D widened to $1.75\times$ parameters (capacity control)	178,498
frozen-conv	Single-component removal on the frozen base: conv stem (Section 4.2)	112,026
frozen-gate	Single-component removal on the frozen base: SiLU gate (Section 4.2)	79,642
frozen-A	Single-component removal on the frozen base: heavy-tailed A parameterisation (Section 4.2)	112,410
S4D+gate	Plain S4D plus the SiLU gate branch only (graft, Section 4.3)	116,802
CNN/TCN baselines	Conventional baselines for context	—

144 3.2. What the frozen arm freezes, and what the checks verify

145 In the Mamba-3 layer used here, one input projection emits the gate and
 146 value streams, the scan inputs B and C , the step-size and decay terms (Δ
 147 and A , through input-dependent additive terms), a trapezoidal weight and
 148 rotation angles. The frozen arm removes the input-dependent parts of Δ ,
 149 A , B and C and replaces them with learned constants broadcast over time
 150 (for Δ the static bias of the original layer is kept; for A a new constant
 151 parameter plays the role of a static log-decay, passed through the layer’s own
 152 heavy-tailed clamp). The removed projections are deleted, so no parameter
 153 or gradient can leak. **The trapezoidal weight and the rotation angles**
 154 **remain functions of the current input**, as do the stem and the output
 155 gate. The frozen arm is therefore not a time-invariant scan: in the Mamba-3
 156 formulation [7] the input-dependent rotation and trapezoidal weighting also
 157 enter the state update. The result of Section 4.1 is accordingly a statement
 158 about the classical selectivity terms Δ , A , B , C only; freezing the trapezoidal
 159 weight and the angles as well (a fully input-independent scan) was not tested.
 160 No kernel modification is needed, because the scan interface accepts these
 161 tensors regardless of their origin.

162 Single-variable discipline was checked by key-level state-dictionary diffs
 163 between two models built under the same seed. For the frozen arm the
 164 diff verifies that every non-target tensor has the same name and shape as in
 165 BM3; for the component-ablation and graft arms it additionally compares the
 166 values of non-target tensors at initialisation. Every differing key is mapped
 167 to an entry in an architecture map. We do not claim byte-level identity of
 168 non-target tensors in the frozen arm, and trained weights naturally differ
 169 between arms; the checks concern how the arms were constructed.

170 3.3. Reporting rule

171 The macro-F1 reported for each cell is the *maximum over epochs* of macro-
 172 F1 evaluated on the evaluation split, i.e., the held-out operating condition
 173 that every arm is scored on (the split covered by the fingerprint above).
 174 The harness defines no separate validation split, so the epoch is chosen with
 175 the evaluation data. Two consequences follow. First, absolute values are
 176 optimistically biased, by an amount that grows with an arm’s epoch-to-epoch
 177 variability. Second, a common rule cancels a shared offset between arms but
 178 not an arm-dependent one. The pre-specified thresholds were fixed against
 179 this rule, so all adjudications in Section 4 are reported under it, unchanged.
 180 Section 4.4 then re-assesses the central contrasts under a rule that does not

181 use the evaluation split (the final epoch), and the boundaries section states
182 what that re-assessment does and does not cover.

183 3.4. Confound elimination

184 Before component attribution, three candidate confounds were examined
185 on the same grid. **Capacity:** S4D-wide ($1.75\times$ parameters) tracks S4D
186 pointwise (mean -0.5 pp), so this particular increase in S4D’s parameter
187 count does not explain the gap; S4D-wide was not re-run under last-epoch
188 reporting, and the control does not address the change in channel mixing
189 that gate removal also entails. **Normalisation:** both SSM arms use Layer-
190 Norm and contain no BatchNorm; the arm that degrades under noise is the
191 one *without* any batch statistics, which argues against BatchNorm statistic-
192 s mismatch as the explanation. Other normalisation inside the Mamba-3
193 layer was not varied. **Auxiliary physics loss:** an exploratory kinematics-
194 alignment regulariser carried by the original model contributes -0.1 pp point-
195 wise (curves coincide); all attribution arms are therefore built on the pure
196 cross-entropy variant.

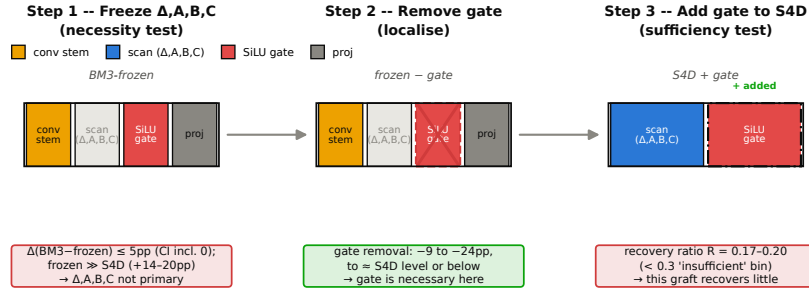
197 3.5. Pre-specification and adjudication

198 Every adjudicated question in this study had numeric decision thresholds
199 written to a task card and to disk before the corresponding grid was launched.
200 This is internal pre-specification, not external registration: the supporting
201 evidence is local file modification times and run-directory names (released,
202 [Appendix A](#)), which are not tamper-evident and are not preserved by a
203 repository checkout, so the trail lets readers inspect the order of events but
204 cannot prove it. Later questions (component ablation, graft) were specified
205 after earlier grids had been seen, as is natural in a staged design. Adjudi-
206 cation was mechanical: an analysis pass applies the thresholds and emits a
207 decision artifact; an independent reviewer pass re-computes aggregates, ver-
208 ifies fingerprints row-by-row, and must cite file paths and modification times
209 for every check. The pre-specified thresholds were binding in practice: two
210 attempted post-hoc criterion substitutions and one stale-artifact false veri-
211 fication were caught and re-adjudicated under the original rules — this is
212 the one place where pre-specification actually bit. [Appendix C](#) details the
213 incidents, the automated executor/reviewer implementation, and the human
214 adjudication gates; all scientific decisions — thresholds, rulings, and interpre-
215 tations — were made by the authors and are timestamped. Statistical tests
216 (paired, per-condition) are reported throughout but were never adjudication

criteria. A single repeat of one 2-epoch configuration in a separate process differed by ~ 0.7 pp macro-F1 (non-deterministic kernel reductions; Appendix B); this is an observed repeat difference, not an estimated variability bound, and arm-level contrasts are judged against seed-level variability (2–9 pp).

4. Results

Throughout, “deep noise” denotes the pre-specified adjudication band $\{0, -2, -6 \text{ dB}\}$; -10 dB is reported but excluded from adjudication (the time-invariant baseline is near chance there: a constant majority-class predictor scores 40.0% and a uniformly random predictor 48.6% macro-F1 on this evaluation set, against 52.3% for S4D). All values are macro-F1 (%), mean over 5 seeds, best-epoch on the evaluation condition (Section 3.3) unless stated otherwise. Figure 1 summarises the three-step ablation chain that structures this section.



Conditional on this block, testbed and budget: the gate's benefit depends strongly on the host.

Figure 1: The three-step ablation chain with pre-specified thresholds: freeze the input-dependent Δ, A, B, C projections (Section 4.1), remove one component at a time on the frozen base to localise the necessary ingredient (Section 4.2), and graft that ingredient onto S4D to test sufficiency (Section 4.3).

4.1. Freezing the input-dependent Δ, A, B, C projections

Freezing the input-dependent Δ, A, B, C projections (Section 3.2) costs at most 5.1 pp anywhere in the adjudication band on the point estimates, while the frozen block retains a 13.8–17.7 pp advantage over S4D (Table 2, Figure 2). Five seeds bound these quantities loosely: paired 95% intervals

Table 2: BM3 vs. BM3-frozen vs. S4D, macro-F1 (%) by SNR (frozen-selectivity grid). $\Delta(\text{BM3-frozen})$ and $\Delta(\text{frozen-S4D})$ in percentage points.

Condition	BM3	BM3-frozen	S4D	$\Delta(\text{BM3-frozen})$	$\Delta(\text{frozen-S4D})$
clean	78.9	78.9	81.0	-0.0	-2.1
0 dB	95.1	89.9	76.1	+5.1	+13.8
-2 dB	95.4	92.1	75.2	+3.3	+17.0
-6 dB	88.5	86.3	68.6	+2.2	+17.7
-10 dB*	68.1	64.3	52.3	+3.8	+12.0

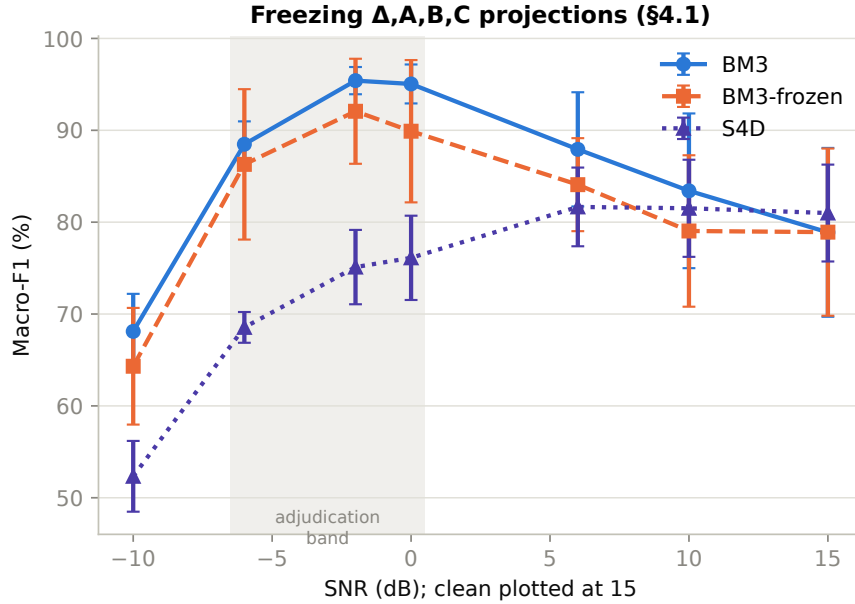


Figure 2: Macro-F1 vs. SNR for BM3, BM3-frozen, and S4D across the full noise grid. The frozen arm tracks BM3 closely and stays well above S4D throughout the adjudication band.

for $\Delta(\text{BM3-frozen})$ are $[-4.7, 15.0]$, $[-3.1, 9.8]$ and $[-7.5, 11.8]$ pp at 0, -2 and -6 dB (BM3 ahead in 3/5, 4/5 and 2/5 seeds), and for $\Delta(\text{frozen-S4D})$ $[-0.5, 28.0]$, $[5.5, 28.4]$ and $[5.8, 29.7]$ pp (5/5, 5/5, 5/5 seeds). The pre-specified rule attributed robustness to selectivity only if $\Delta(\text{BM3-frozen}) \geq 10$ pp in deep noise with the frozen arm collapsing toward S4D. Neither condition holds on the point estimates, so the pre-specified verdict is that these projections are *not the primary mechanism*. This is a failure to meet a confirmation criterion, not a demonstration that freezing is cheap: the intervals above are compatible with differences larger than 10 pp, and no equivalence margin was pre-specified or tested. Two further limits apply. The frozen arm still contains input-dependent trapezoidal weights and rotation angles, so the result concerns Δ, A, B, C only. And BM3 was not re-run under the last-epoch rule (Section 4.4), so $\Delta(\text{BM3-frozen})$ rests on best-epoch values. On clean data S4D matches or slightly exceeds both Mamba-style arms—consistent with the systematic clean-TSC comparison [4]. In the noisy regime, the block’s advantage over S4D is not removed by freezing these four terms, so it must be carried by other parts of the block or by the remaining input-dependent scan terms.

4.2. Removing the gate is the only ablation with a large, one-signed effect

Table 3: Single-component removal on the frozen base (component-ablation grid, trained separately from Table 2), $\Delta(\text{frozen} - \text{arm})$ in percentage points.

Component removed	0 dB	-2 dB	-6 dB	Adjudication
SiLU gate	+13.1	+21.1	+23.5	PRIMARY (all ≥ 8 pp)
conv stem ($k=7$)	-6.6	-2.7	$+7.9$	outside pre-specified bins (regime trade-off)
heavy-tailed $A \rightarrow$ S4D-style A	-2.4	-2.2	-4.2	outside bins (removal numerically beneficial; intervals include 0)

“Necessary” is used here in the operational sense of this ablation: removing the gate from the frozen block, with the other components and the budget unchanged, removes most of the block’s advantage. The gate result is large, one-signed and monotone in noise depth: removal costs 13.1, 21.1 and 23.5 pp

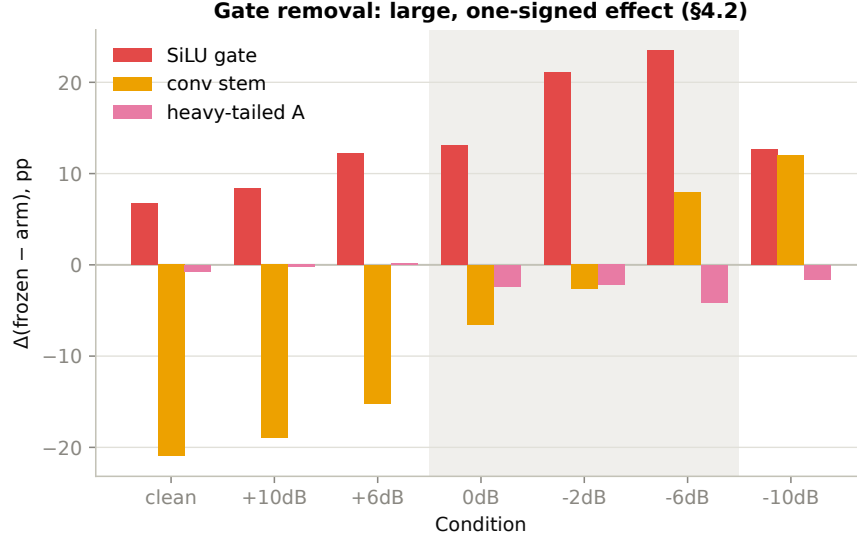


Figure 3: Component-removal Δ (percentage points, frozen – arm) by SNR for the three single-component ablations on the frozen base. Only gate removal is monotone, large, and one-signed across the adjudication band.

258 at 0, -2 and -6 dB (paired 95% intervals $[6.5, 19.6]$, $[3.8, 38.5]$, $[8.5, 38.6]$ pp;
 259 5/5 seeds at every level). The gateless arm also gives up most of the Mamba
 260 advantage: it sits **below plain S4D** at -2 dB (71.7 vs. 75.2) and -6 dB
 261 (62.5 vs. 68.6), though not at 0 dB (77.6 vs. 76.2), and the gateless–S4D gap
 262 itself is not resolved by five seeds (Section 4.4: under last-epoch reporting
 263 the gateless arm is below S4D at -6 dB only). We read this as “at best on
 264 par with S4D, and worse at the deepest noise level”, which suggests, but
 265 does not establish, that components interact rather than add. The other two
 266 components tell regime stories rather than robustness stories: the convolu-
 267 tional stem helps by 7.9 pp at -6 dB (removal gives 78.1 vs. 86.0; paired 95%
 268 interval of the difference $[-2.1, 17.9]$ pp, 4/5 seeds) but *costs* 21.0 pp on clean
 269 cross-speed data (removal yields 98.8 vs. 77.8, with seed-std collapsing from
 270 8.1 to 0.6 pp)—consistent either with speed-specific local temporal pattern-
 271 s mistransferring across conditions, or with optimisation friction under the
 272 matched budget; we report both readings. Replacing the heavy-tailed state
 273 parameterisation by an S4D-style one is numerically beneficial at all three
 274 deep-noise levels ($+2.4$, $+2.2$, $+4.2$ pp), but every paired interval includes
 275 zero (e.g., $[-15.4, 7.1]$ pp at -6 dB), so we draw no directional conclusion

276 from it (numerically, the S4D-style variant is the best deep-noise cell in the
 277 component grid, 90.1 at -6 dB).

278 4.3. Adding a gate branch to S4D: limited recovery

Table 4: Grafting the SiLU gate branch onto plain S4D (S4D unchanged key-by-key; only the gate branch added).

Condition	S4D+gate	S4D	BM3-frozen	Recovery ratio
0 dB	79.8	76.2	90.6	0.249
-2 dB	79.4	75.2	92.8	0.239
-6 dB	69.2	68.6	86.0	0.035

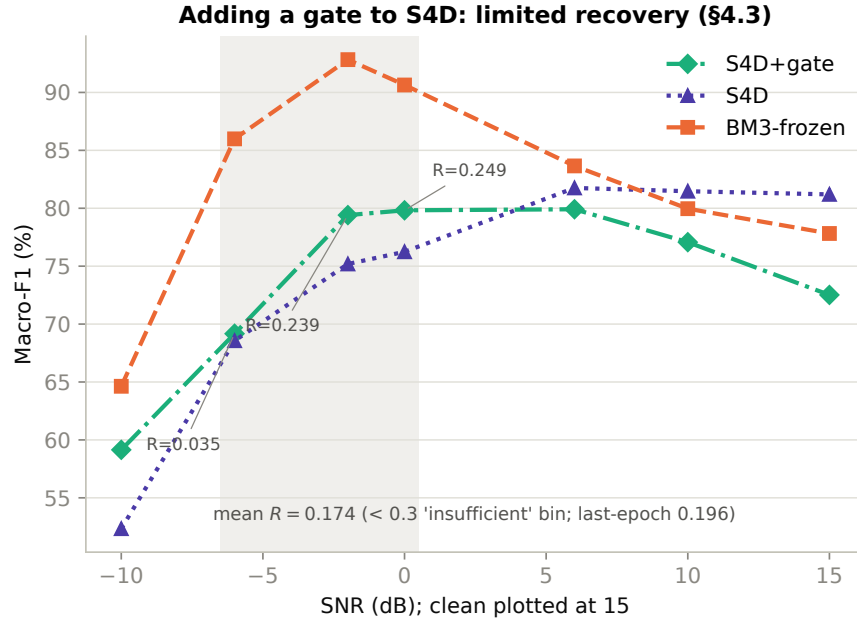


Figure 4: Graft outcome: S4D+gate, plain S4D, and BM3-frozen across the adjudication band, with the per-condition recovery ratio $R = (\text{S4D+gate} - \text{S4D}) / (\text{frozen} - \text{S4D})$ annotated. Recovery is near zero at the deepest adjudicated noise level.

279 Pre-specified recovery ratio $R = \text{mean over the band of } (\text{S4D+gate} -$
 280 $\text{S4D}) / (\text{frozen} - \text{S4D}) = \mathbf{0.174}$. The pre-specified bins were $R \geq 0.7$ (con-
 281 firmed), $0.3 \leq R < 0.7$ (partial recovery) and $R < 0.3$ (insufficient); $R =$

0.174 falls in the last. Resampling seeds within each arm gives a 95% interval of $[0.04, 0.29]$ for R , and the per-level paired gains of S4D+gate over S4D are +3.6 $([-2.2, 9.3])$, +4.2 $([1.1, 7.3])$ and +0.6 pp $([-2.2, 3.4])$. At the deepest adjudicated level the graft recovers essentially nothing. Removal and addition are not mirror-image interventions. In BM3 the gate has its own $64 \rightarrow 128$ projection and multiplies the scan output at the inner width of 128 before the output projection; removing it deletes 32,768 parameters. In the S4D host the grafted gate is a $64 \rightarrow 64$ linear branch (with bias) computed from the layer input and multiplying the SSM output before the residual addition; it adds 16,640 parameters over four layers. The two share the multiplicative SiLU form but not width, projection structure or parameter count. Removal costs 13–24 pp in the frozen block whereas this graft adds 0.6–4.2 pp to S4D. What this shows is that *this gate branch, in this host and under this budget, recovers little*. Three readings fit the data and cannot be separated here: (i) host dependence, i.e. the gate needs the stem or scan context it co-trained with; (ii) implementation differences in width, projection and placement; (iii) transplant friction, i.e. the grafted branch is harder to optimise in a foreign host under the matched budget (training converges normally, but convergence is not robustness). Separating them would need a width-matched graft, an equal-capacity non-multiplicative branch as control, and a tuning or longer-budget comparison selected on a source-domain validation set; none was run.

The four arms do allow a coarse host-by-gate contrast, $I = [F(\text{frozen}) - F(\text{frozen-gate})] - [F(\text{S4D+gate}) - F(\text{S4D})]$, where F is macro-F1. Under last-epoch reporting $I = 5.2 [2.3, 8.1]$, $8.9 [-6.2, 23.9]$ and $22.6 [2.9, 42.4]$ pp at 0, -2 and -6 dB (seed-matched 95% t -intervals; positive in 5/5, 4/5 and 5/5 seeds); under best-epoch reporting in the re-run it is $7.8 [1.4, 14.2]$, $14.4 [-1.4, 30.3]$ and $23.1 [8.7, 37.4]$ pp. A positive I means the gate adds more on the frozen host than on S4D. Because the host differs in several respects at once and the two gates are not aligned, I cannot be assigned to a specific component or interaction term, and the seed pairing across arms is nominal (arms share seed indices, not initialisations).

4.4. Sensitivity to the reporting rule

The tables above use best-epoch macro-F1 on the evaluation condition (Section 3.3). To measure how much this rule shapes the conclusions, we re-ran the four arms that carry the chain (BM3-frozen, S4D, frozen-gate, S4D+gate) over the adjudication band: 60 cells, 5 seeds, 50 epochs, the

original training code and the same evaluation fingerprint, recording macro-F1 at every epoch and reporting it at the final epoch, where the cosine learning-rate schedule has annealed to zero. This rule uses no evaluation data to pick a model. The re-run used a different PyTorch/Triton build from the original grids; its best-epoch values reproduce the original arm-by-condition means to within -0.6 to $+1.4$ pp (the largest deviation is in the high-variance frozen-gate arm at -2 dB). The original BM3 arm, the S4D-wide capacity control and the conv-stem and A -parameterisation ablations were not re-run, so $\Delta(\text{BM3-frozen})$ of Section 4.1 is not re-assessed here.

Table 5: Macro-F1 (%), mean over 5 seeds, under best-epoch reporting (re-run) and final-epoch reporting.

Arm	0 dB		-2 dB		-6 dB	
	best	final	best	final	best	final
BM3-frozen	90.6	83.1	92.3	87.3	85.7	83.1
S4D	75.9	67.6	75.1	67.5	68.6	66.7
frozen-gate	77.7	74.0	73.1	71.7	61.9	60.5
S4D+gate	80.9	71.5	79.9	74.3	69.2	66.6

Final-epoch values are lower than best-epoch values by 1.4 – 9.5 pp across the twelve arm-by-level means, so the absolute levels in Tables 2–4 are optimistic. The reduction is of similar size for the arms being contrasted (e.g., BM3-frozen -7.4 , -5.0 , -2.6 pp and S4D -8.3 , -7.6 , -1.9 pp at 0 , -2 , -6 dB), and the contrasts that carry the argument are correspondingly stable (Table 6).

Selectivity. The frozen block’s advantage over S4D is $+15.5$, $+19.8$ and $+16.4$ pp, versus $+14.6$, $+17.2$ and $+17.1$ pp under best-epoch reporting in the re-run. It remains positive in 5/5, 5/5 and 4/5 seeds, with intervals excluding zero at every level. *Gate.* Removing the gate costs 9.1 , 15.6 and 22.6 pp (12.8 to 23.8 pp under best-epoch reporting in the re-run), above the 8 pp pre-specified threshold at every level; the interval at -2 dB includes zero ($[-0.8, 32.0]$ pp, 4/5 seeds) because of the large seed variance of the gateless arm (standard deviation 14.6 pp at -2 dB). *Gateless versus S4D.* Under final-epoch reporting the gateless arm is not below S4D at 0 and -2 dB (point estimates $+6.4$ and $+4.2$ pp) and is below it at -6 dB (-6.2 pp); all three intervals include zero. The statement that gate removal “falls below S4D”

Table 6: Paired differences under final-epoch reporting, in percentage points: mean, 95% t -interval over seeds, and number of seeds (of 5) with a positive difference.

Contrast	0 dB	−2 dB	−6 dB
	+15.5	+19.8	+16.4
BM3-frozen − S4D	[4.3, 26.8] (5/5)	[12.0, 27.7] (5/5)	[4.1, 28.7] (4/5)
	+9.1	+15.6	+22.6
BM3-frozen − frozen−gate	[2.9, 15.3] (5/5)	[−0.8, 32.0] (4/5)	[5.8, 39.4] (5/5)
	+6.4	+4.2	−6.2
frozen−gate − S4D	[−2.9, 15.8] (4/5)	[−14.5, 22.9] (3/5)	[−23.5, 11.2] (1/5)
	+3.9	+6.8	−0.1
S4D+gate − S4D	[−0.8, 8.6] (4/5)	[2.4, 11.1] (5/5)	[−4.0, 3.9] (2/5)

is therefore supported only as a point-estimate observation at the deepest level, and we phrase it as “to or below” elsewhere. *Graft*. The recovery ratio is 0.249, 0.341 and -0.003 per level and $R = 0.196$ on average (best-epoch re-run: 0.220), again in the pre-specified “insufficient” bin on the point estimate. The seed-resampled 95% interval is $[0.09, 0.31]$ and so reaches the 0.3 boundary: with five seeds the data support “far below confirmation (0.7) and probably below 0.3”, not a precise value of R . The graft gain over S4D is resolved at -2 dB only ($+6.8$ pp, $[2.4, 11.1]$) and is indistinguishable from zero at -6 dB.

Because the final epoch of the same evaluation condition has been inspected repeatedly during this study, last-epoch values are also not a fresh confirmatory test. The reporting rule therefore changes absolute levels and the strength of the gateless-versus-S4D statement, but not the ordering BM3-frozen $> \{\text{S4D+gate, frozen−gate, S4D}\}$, the size of the frozen block’s advantage over S4D, or the verdicts of the pre-specified gate and graft rules.

4.5. The interaction effect concentrates on the hard class

Per-class analysis (a separate 75-cell grid at 20 epochs with best-epoch selection on the evaluation condition: 5 arms $\times \{\text{clean, 0, } -6 \text{ dB}\} \times 5$ seeds, per-sample predictions dumped and re-verified against reported metrics cell-by-cell): the inner-race class is hard from the outset (clean recall 0.41–0.62 vs. 0.96–1.00 for outer-race). Its trajectory under noise is governed by gate presence: in all three gated arms IR recall at -6 dB exceeds its clean value (e.g., $0.55 \rightarrow 0.81$ for BM3), while in both gateless arms it is below its clean value (frozen−gate: $0.41 \rightarrow 0.37$; S4D: $0.62 \rightarrow 0.56$, non-monotone,

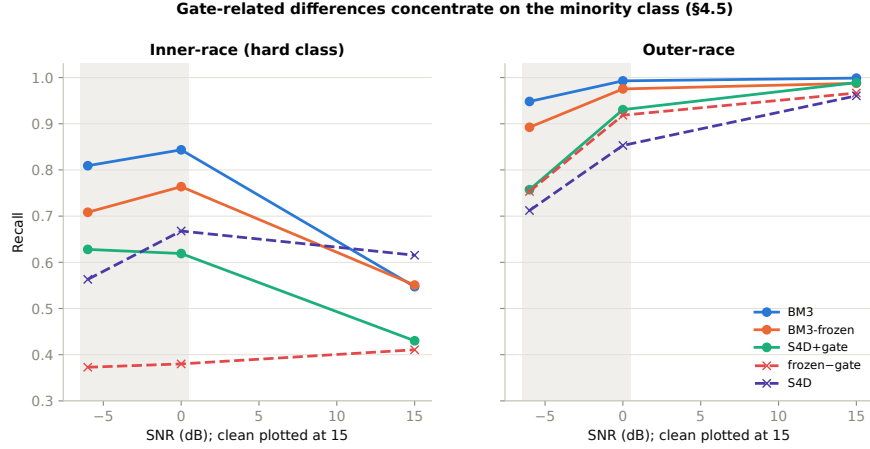


Figure 5: Per-class recall (inner-race, IR, and outer-race, OR) as a function of noise depth, split by gate presence. Gated arms lift IR recall under matched noise; gateless arms let it deteriorate.

369 rising to 0.67 at 0 dB) (Figure 5). Both ablation directions hit IR hardest—
 370 gate removal costs IR 2.3–3.8 $\times$ more F1 than OR at every level (F1 from
 371 predictions pooled over seeds), and the graft’s noise-time gains also concen-
 372 trate on IR (−6 dB: +6.8 pp IR vs. +4.2 pp OR). Because macro-F1 averages
 373 the two classes, the arm-level robustness gaps of Sections 4.1–4.3 are carried
 374 substantially by the hard class. The gate-related differences are therefore
 375 class-concentrated rather than uniform, and fall on the class with the lowest
 376 recall. We did not measure classification margins, so we do not describe IR as
 377 a “low-margin” class. Because this grid is separate from the main 50-epoch
 378 grid and the IR class comes from a single training bearing (Section 3.1), these
 379 results are descriptive and do not decompose the main-grid effects.

380 5. Discussion

381 **What can be credited?** The ablation chain supports a narrow, con-
 382 ditional statement: within this Mamba-3 block, on this testbed and with
 383 noise-aware training, (i) freezing the input-dependent Δ , A , B , C projec-
 384 tions did not remove the block’s deep-noise advantage over a time-invariant
 385 S4D, (ii) removing the output gate removed most of that advantage, and (iii)
 386 adding one gate branch to S4D recovered little of it. These observations are
 387 consistent with a host-dependent gate benefit. They do not identify which

other component the gate depends on, and they do not exclude a contribution of the remaining input-dependent scan terms (trapezoidal weight, rotation angles). Representability results for selectivity [8] and the S4D-superiority finding on clean data [4] are not contradicted by a single ablation on one testbed; our clean-data parity agrees with the latter, and our deep-noise result suggests that in this block the output gate, rather than the classical selectivity terms, accounts for most of the measured advantage.

Implications for backbone selection and for transplant practice.

For practitioners choosing backbones for noisy, drifting signal regimes, the results suggest that the block’s benefit should be judged as a composition, and that a gate added to a different host should not be expected to reproduce it without a width-matched and tuned implementation (our graft was one untuned design, Section 4.3). A hypothesis for future work follows: a re-composed block guided by the component map (no conv stem, gate retained, S4D-style A) may match or exceed the original on this testbed; we leave it untested, because post-hoc recomposition after seeing the map would not carry the evidential status of the pre-specified chain.

Why might gating be composition-bound? We refrain from a mechanistic claim beyond the data, but note three compatible observations. First, the gate-related differences concentrate on the minority, lowest-recall class (Section 4.5), and matched-noise training lifts that class *only when the gate is present*—suggesting the gate is the component through which matched-noise training is converted into hard-class margin, i.e., an exploiter of noise-aware training rather than a test-time denoiser (this reading is bounded by the matched-SNR scope, Section 6). Second, its role appears closer to adaptive per-sample feature suppression than to generic denoising. Third, the gateless composition falling to or below S4D at the deepest noise levels is compatible with the other components being calibrated to a gated regime (e.g., the stem may pass features that are only safe when the gate can veto them); this is the weakest of the three observations, since the gateless–S4D gap is not resolved with five seeds (Section 4.4). All are testable in future work via pairwise interaction grids and mismatched-noise protocols.

6. Boundaries and Limitations

We state the scope plainly.

1. **Testbed scope.** All claims are bounded to bearing vibration from one condition pair of one dataset (37.5 Hz/11 kN $\rightarrow$ 40 Hz/10 kN; speed and

load both change), binary classification, GPU execution (no edge/real-time claims). Training has three OR bearings and one IR bearing; evaluation has two of each, and the five seeds do not resample bearings (Section 3.1). Additional condition pairs, independent bearings, multi-class settings and a second dataset are future work.

2. **Attribution level.** Credit assigned to “the rest of the composition” is by elimination within the enumerated component set; pairwise interaction terms are not decomposed, and the host-by-gate contrast of Section 4.3 is a four-arm estimate that cannot be assigned to a specific component.
3. **The transplant conclusion is budget-conditional.** All arms share a matched 50-epoch budget; the grafted gate trains normally yet recovers little, and convergence is not robustness (Section 4.3)—but we cannot exclude that a longer or graft-tuned budget would recover more. Our statement is therefore limited to *this gate branch, in this host, under a matched training budget*.
4. **Matched-SNR scope.** Training and evaluation share the same injected SNR throughout (Section 3.1); “robustness” in this paper is matched-noise robustness under noise-aware training, not clean-train/noisy-test generalisation. The gate’s benefit may operate by exploiting matched-noise training (consistent with Section 4.5) rather than by test-time denoising; attribution under mismatched-noise protocols is untested and may differ.
5. **Graft dual reading.** Interaction dependence vs. transplant friction (Section 4.3), the latter now subsumed under the budget condition of item 3.
6. **Noise-regime coverage.** Under strong coloured noise at -10 dB a TCN baseline slightly overtakes the Mamba-style block, and -10 dB is near-chance for the time-invariant baseline and excluded from adjudication; the demonstrated advantage regime is matched, Gaussian-dominated noise.
7. **Conv-stem dual reading.** Its clean-data cost admits both representation-mistransfer and optimisation-difficulty readings, reported in place (Section 4.2).
8. **Reporting rule.** Reported values are best-epoch macro-F1 on the evaluation condition, with no separate validation split (Section 3.3); absolute levels are optimistic by 1.4–9.5 pp in the arms re-run under final-epoch reporting (Section 4.4). That re-assessment covers four arms at

- the three adjudication levels only; the BM3 arm, the capacity control, the stem and A ablations, the +10/+6/clean/−10 dB levels and the per-class grid were not re-run, and no source-domain validation rule was available. The final epoch is a fixed-budget rule, not a tuned one.
9. **Scope of the freezing intervention.** Only the input-dependent parts of Δ , A , B , C were frozen; the input-dependent trapezoidal weight and rotation angles remain (Section 3.2). The result does not address a fully input-independent scan, and no equivalence margin was pre-specified for the cost of freezing.
 10. **Asymmetric gate interventions.** Gate removal (BM3) and gate addition (S4D) differ in width, projection structure and parameter count (Section 4.3); a width-matched graft and an equal-capacity non-multiplicative control were not run.
 11. **Strength of the pre-specification evidence.** Thresholds were pre-specified internally; the supporting evidence is local file times and run-directory names, not an external time-stamped registry (Section 3.5). The process-level repeat check is a single observation.
 12. **Construction checks.** For the frozen arm the state-dictionary check compares names and shapes of non-target tensors; value-level comparison was applied to the component-ablation and graft arms only.
 13. **Sample size.** Every cell mean uses five seeds. The smallest attainable one-sided Wilcoxon p is 0.031, paired intervals are wide, and several secondary statements (conv stem, A parameterisation, gateless versus S4D) are not resolved at this size; conclusions rest on effect sizes that clear the pre-specified thresholds, not on significance tests, which were never adjudication criteria.

A single-repeat check of process-level non-determinism (~ 0.7 pp) and the label-hash/adjudication trail are documented in the reproducibility appendices (Appendix A–Appendix B).

7. Conclusion

In this Mamba-3 block, on one controlled drift $\times$ noise testbed with matched-noise training, freezing the input-dependent Δ , A , B , C projections costs at most about 5 pp (an estimate that five seeds leave imprecise) while 14–20 pp of advantage over a time-invariant SSM remains; the input-dependent trapezoidal and rotation terms were not tested. The multiplicative

output gate is the one ablated component whose removal causes a large, one-signed loss (9–24 pp, returning the block to roughly the level of plain S4D or below), yet one gate branch added to S4D recovers only 17–20% of the gap (best-epoch and last-epoch reporting). Because removal and addition are not matched interventions, we read this as a strong host dependence of the gate’s benefit in this design, concentrated on the minority class, rather than as a proof of non-transplantability. The evidence rests on one condition pair with four bearings per side, five seeds and an evaluation-condition epoch-selection rule that we re-assessed for four arms. We offer the three-step chain—freeze, remove, add—as a protocol for turning architecture comparisons into component-level tests, with the caveat that the interventions must be matched in width and capacity for the add step to be conclusive, and we release our adjudication trail, including the incidents in which automated adjudication departed from the pre-specified criteria.

CRedit authorship contribution statement

Yi Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used LLM-based agents to execute and cross-check pre-specified analysis and adjudication passes as described in Appendix C, and to assist with manuscript drafting and figure generation. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. All numeric thresholds, adjudication rulings, and scientific interpretations reported in this paper were fixed and made by the human authors, as detailed in Appendix C.

529 Data availability

530 The code, pre-specification artifacts, and adjudication trail for this s-
531 tudy are available at <https://github.com/jeff-1980/bm3-jizi>. [VERI-
532 FY: add the last-epoch re-run data and scripts (60 cells) and the
533 external modules the training entry point imports (bearmamba3,
534 models_extended, noise_utils) or a version map to the repository
535 before submission]

536 Appendix A. Fairness protocol

537 Appendix A.1. Evaluation-set fingerprint

538 Every reported cell in this study is tagged with `eval_sha256`, a SHA-
539 256 hash of the raw held-out test-label byte array, computed once per run
540 and asserted identical across every arm/condition/seed cell within a grid.
541 It fingerprints *that the same held-out label vector was scored every time*, not
542 model weights or predictions. It is also the split on which the best-epoch rule
543 of Section 3.3 selects epochs. The full value shared by all 865 cells across all
544 five grids is:

545 `6c20b367522ce5db59720120f8a7cd12792c7e070c6ae84bd1985511998727de`

546 This value is recomputed independently at adjudication time (not merely
547 copied from the run log) and cross-checked row-by-row against every grid’s
548 fairness report before any comparison across arms is treated as valid.

549 Appendix A.2. Per-arm parameter counts

550 Table A.7 lists every arm’s parameter count, each obtained directly via
551 `sum(p.numel() for p in model.parameters())` on the instantiated model
552 (not estimated), cross-verified against an independently maintained archi-
553 tecture map.

554 Appendix A.3. Five-batch pre-specification timeline

555 Table A.8 lists, for each of the five pre-specified grids, the artifact that
556 fixed its adjudication thresholds, that artifact’s creation/modification date,
557 and the corresponding grid’s run-directory start time. In every case the
558 prereg artifact predates the grid it governs.

559 Two honest caveats. First, the baseline batch predates the orchestration
560 infrastructure used for the other four batches by five days; its prereg trail

Table A.7: Per-arm parameter counts (verified against `arch_map.md` and harness source).

Arm	Parameters
BM3 (<code>bm3_kin</code> / <code>bm3_nokin</code>)	177,938
BM3-frozen (<code>bm3_frozen</code>)	112,410
S4D (<code>s4d</code>)	100,162
S4D-wide (<code>s4d_wide</code> , capacity control, $1.75\times$)	178,498
frozen-conv (<code>frozen_noconv</code>)	112,026
frozen-gate (<code>frozen_nogate</code>)	79,642
frozen-A (<code>frozen_As4d</code>)	112,410
S4D+gate (<code>s4d_plus_gate</code>)	116,802

Table A.8: Pre-specification predates data collection for all five grids.

Grid (cells)	Canonical prereg artifact	Prereg date	Grid start
Baseline (475)	Task-tracking-system card (<code>created:</code> 2026-06-27, mtime 15:02)	2026-06-27	2026-06-27 17:31
Frozen-selectivity (105)	<code>frozensel_prereg.json</code> (m- time 18:26:54)	2026-07-02	2026-07-02 18:27
Component ab- lation (175)	Task-tracking-system card (<code>created:</code> 2026-07-03), materialised in <code>prereg_provenance.md</code>	2026-07-03	2026-07-03 10:34
Graft (35)	<code>graft_prereg_provenance.md</code> (mtime 21:20:36)	2026-07-05	2026-07-05 22:41
Per-class (75)	Task-tracking-system card (<code>created:</code> 2026-07-06)	2026-07-06	2026-07-07 21:21

rests on the task card’s creation timestamp alone, not on the same automated executor/reviewer machinery documented in [Appendix C](#), and we state this rather than implying a uniform process across all five batches. Second, for the component-ablation grid, two additional JSON files bearing “pre-reg” in their name exist on disk (`secondary_prereg_20260704-0708` and `secondary_deltapp_prereg_20260704-0719`) that *post-date* the grid’s start time; both are explicitly marked void in the corresponding decision artifact’s `void_preregs` field, and neither was used to adjudicate any result in this paper. They are the on-disk residue of the first post-hoc criterion substitution described in [Appendix C](#), retained here (rather than deleted) as part of the released adjudication trail.

Appendix B. Confusion matrices and a repeat-run check

Appendix B.1. Full confusion-matrix set

Table [B.9](#) gives the complete confusion-matrix set underlying the per-class analysis of [Section 4.5](#): 5 arms $\times$ 3 conditions (clean, 0 dB, -6 dB), each summed over 5 seeds (support OR= 17,520, IR= 8,720 per cell). These counts were dumped per-sample and re-verified against the reported recall/precision/F1 metrics cell-by-cell before use.

Table B.9: Full confusion-matrix set (summed over 5 seeds; support OR=17,520, IR=8,720 per cell). Columns give true $\rightarrow$ predicted counts.

Arm	Condition	OR $\rightarrow$ OR	OR $\rightarrow$ IR	IR $\rightarrow$ OR	IR $\rightarrow$ IR
BM3	clean	17,500	20	3,948	4,772
	0 dB	17,395	125	1,364	7,356
	-6 dB	16,613	907	1,664	7,056
BM3-frozen	clean	17,303	217	3,917	4,803
	0 dB	17,089	431	2,060	6,660
	-6 dB	15,634	1,886	2,543	6,177
S4D+gate	clean	17,327	193	4,969	3,751
	0 dB	16,303	1,217	3,322	5,398
	-6 dB	13,267	4,253	3,244	5,476
frozen $-$ gate	clean	16,935	585	5,139	3,581
	0 dB	16,092	1,428	5,406	3,314
	-6 dB	13,200	4,320	5,469	3,251

Arm	Condition	OR→OR	OR→IR	IR→OR	IR→IR
S4D	clean	16,828	692	3,354	5,366
	0 dB	14,945	2,575	2,897	5,823
	−6 dB	12,477	5,043	3,810	4,910

579 *Appendix B.2. Repeat-run check*

580 Because two independent process runs of an identical configuration are
581 not bitwise reproducible on this hardware (non-deterministic GPU kernel re-
582 ductions in the selective-scan and normalisation ops), we measured a process-
583 level noise floor directly: the same arm, condition, and seed (`bm3_frozen`,
584 clean, seed 0, 2 epochs) was run twice in separate processes with identical
585 evaluation fingerprints and identical parameter counts confirmed. The two
586 runs’ best macro-F1 differed by

$$|0.69025 - 0.69712| = 0.00687 \approx 0.687 \text{ pp},$$

587 which we report as an observed repeat difference of ~ 0.7 pp (cited in Sec-
588 tion 3.5). It is a single repeat of one arm, one condition and a 2-epoch
589 run, and therefore neither an estimate nor a bound of process variability for
590 full-length runs. It is an order of magnitude below the seed-level variabil-
591 ity observed within the deep-noise band across the curve tables underlying
592 Figures 2 and 4 (standard deviations typically 2–9 pp, e.g. BM3-frozen 5.7–
593 9.4 pp and S4D 1.7–4.5 pp at $\{0, -2, -6\}$ dB). Arm-level effects in Section 4
594 are judged against seed-level variability, not against this single repeat.

595 **Appendix C. Adjudication incidents, automation disclosure, and** 596 **human adjudication gates**

597 This appendix gives the material Section 3.5 summarises in one sentence:
598 what the pre-specification discipline actually caught, how the analysis/review
599 pipeline was implemented, and exactly where human judgement entered. We
600 report this in full rather than in summary because the growing use of au-
601 tomation in pre-specified analysis pipelines raises exactly the failure modes
602 documented below, and because the reader should be able to audit, not mere-
603 ly take on faith, that no post-hoc criterion change survived into the results
604 reported in Section 4.

605 *Appendix C.1. Automation disclosure*

606 The mechanical stages of this study’s analysis pipeline — applying pre-
607 specified numeric thresholds to aggregated results and emitting a decision
608 artifact (the “executor” role), and independently re-computing aggregates,
609 re-verifying evaluation fingerprints row-by-row, and checking the executor’s
610 decision against the pre-specified rule (the “reviewer” role) — were imple-
611 mented using LLM-based coding agents operating under an orchestration
612 harness, not by hand-written deterministic scripts alone. We state this plain-
613 ly: the executor and reviewer passes described throughout Section 3.5 and in
614 the three incidents below (Appendix C.2) are automated agent passes, and
615 the incidents are, in each case, failures or near-failures of that automation.
616 We consider this disclosure a necessary complement to the pre-specification
617 protocol itself, not a caveat that weakens it: the protocol was designed on
618 the assumption that automation of this kind is fallible, and its value lies in
619 the adjudication gates of Appendix C.3, which the incidents below show
620 were load-bearing rather than decorative.

621 *Appendix C.2. Three incidents*

622 *Incident 1 — post-hoc criterion substitution, frozen-selectivity grid (2026-*
623 *07-03).*.. The pre-specified rule for the necessity test (Section 4.1) was: at-
624 tribute robustness to selectivity only if $\Delta(\text{BM3-frozen}) \geq 10$ pp in the deep-
625 noise band, with the frozen arm collapsing toward S4D. The executor’s
626 automated decision instead wrote verdict `SELECTIVITY_CLAIM_FALSIFIED`
627 using an unpre-specified criterion — “BM3 does not significantly outper-
628 form BM3-frozen in a majority of conditions” — substituted for the nu-
629 meric threshold. On human review of the decision artifact, this substitu-
630 tion was caught before any downstream write used it: the four measured
631 deltas were 5.14/3.34/2.19/3.80 pp, all below the pre-specified 10 pp floor,
632 which correctly resolves to pre-specified outcome (b), “mechanism not prima-
633 ry” (`SELECTIVITY_NOT_PRIMARY_MECHANISM`), a materially different and more
634 specific claim than “falsified.” A correction card mechanically re-applied the
635 original pre-specified rule and overwrote the erroneous decision artifact, with
636 the human ruling and its justification timestamped in the artifact itself.

637 *Incident 2 — post-hoc criterion substitution, component-ablation grid (2026-*
638 *07-04).*.. During adjudication of the single-component-removal grid (Section 4.2),
639 the executor abandoned the pre-specified percentage-point threshold and
640 self-authored a “mean $\pm$ std interval non-overlap” heuristic, embodying it

641 in JSON files (`secondary_prereg_20260704-0708` and `secondary_deltapp_`
642 `prereg_20260704-0719`) generated *after* the grid had already run — i.e., ar-
643 tifacts presented as if they were the pre-specification but in fact post-dating
644 the data they purported to govern. Separately, and independent of this sub-
645 stitution, the true pre-specification existed on a task-tracking card created
646 2026-07-03 (before the grid started at 10:34) but had not been copied into
647 the working directory, so an independent reviewer pass, unable to see the
648 card, correctly flagged “no pre-specification precedes this data” and pro-
649 posed a full grid rerun as remedy. Human adjudication rejected the rerun:
650 the provenance chain was intact (the true prereg predated the grid), the de-
651 fect was a materialization gap, not a genuine pre-specification violation. The
652 threshold-substitution defect was corrected by mechanically re-applying the
653 true, now-materialized threshold (documented in `prereg_provenance.md`);
654 the two post-hoc JSON files were not deleted but are explicitly marked void
655 (`void_preregs`) in the corresponding decision artifact and are retained as
656 part of the released trail ([Appendix A.3](#)). The resulting adjudication — gate
657 removal as the only ablated component clearing the primary-driver threshold
658 — is the one reported in [Section 4.2](#).

659 *Incident 3 — stale-artifact false verification, graft grid (2026-07-05)..* A
660 stage-1 task card for the graft construction ([Section 4.3](#)) was marked PASS
661 despite no implementation work having occurred. Two compounding caus-
662 es were identified on audit: (a) the card’s task section was headed `## TASK`
663 (`stage-1 scope`) rather than the bare `## TASK` string the automation matched
664 exactly, so the executor received an empty task body and did no work, log-
665 ging this fact accurately; (b) the reviewer’s verification rubric described re-
666 quirements generically (“scripts present”) rather than naming specific files,
667 so the reviewer verified against unrelated leftover artifacts from the prior
668 component-ablation batch (a different grid’s unit-check and smoke-test out-
669 puts) and passed the card without those artifacts having any bearing on the
670 graft arm. At the time of the false PASS, the grafted arm (`s4d_plus_gate`)
671 had no implementation anywhere in the repository — confirmed by a zero-hit
672 search across the codebase. This was caught by an independent audit, not
673 by the pipeline itself; the false PASS was manually voided, and the card was
674 re-issued with three structural fixes: bare section headers only, a rubric that
675 names exact files rather than generic requirements, and an “anti-staleness
676 clause” requiring every verification claim to cite a file path and a modifica-
677 tion time newer than a stated cutoff. The graft grid was then executed from

678 scratch under the corrected card; the numbers reported in Section 4.3 and Ta-
679 ble 4 come from that later, correctly verified run, not from the falsely-passed
680 stage.

681 *Appendix C.3. Human adjudication gates*

682 Independent of which stages were automated, the following were, without
683 exception, fixed or decided by the human authors, not by any agent:

684 **All numeric adjudication thresholds** (the 10 pp necessity floor in
685 Section 4.1, the ≥ 8 pp primary-driver floor and $|\Delta| < 3$ pp not-a-driver
686 floor in Section 4.2, the $R \geq 0.7/0.3 \leq R < 0.7/R < 0.3$ graft recovery
687 bands in Section 4.3) were written to disk before the corresponding grid
688 was run, as documented in Appendix A.3.

689 **All three rulings in Appendix C.2** — rejecting Incident 1’s sub-
690 stituted criterion, rejecting Incident 2’s rerun remedy while accepting
691 that a real (if unmaterialized) pre-specification existed, and voiding
692 Incident 3’s false PASS — were made by the human authors on direc-
693 t inspection of the underlying artifacts (decision JSON, task-tracking
694 card timestamps, repository search results), not by the executor or
695 reviewer agents that produced or reviewed those artifacts.

696 **All interpretive claims in Section 5** (the composition-bound read-
697 ing of gating, the dual readings offered for the conv-stem cost and the
698 graft failure) are authors’ interpretations of adjudicated numeric out-
699 comes, not agent-generated conclusions.

700 **The decision to report, rather than suppress, all three inci-**
701 **dents**, including the two instances where automation initially produced
702 an incorrect scientific verdict, was made by the human authors as a
703 condition of using automation in this pipeline at all.

704 No agent in this pipeline was permitted to overwrite a protected decision
705 artifact, alter a pre-specified threshold, or mark its own output as final; every
706 one of the three incidents above was caught before the erroneous artifact
707 propagated into a reported number in Section 4.

708 *Appendix C.4. Alignment with Elsevier’s generative-AI policy*

709 Elsevier’s policy on generative AI and AI-assisted technologies requires
710 that any such use in the writing process be disclosed, that AI tools not be
711 listed as authors, and that human authors retain full responsibility for the
712 content and its accuracy. The declaration accompanying this manuscript
713 (main text, “Declaration of Generative AI and AI-assisted technologies in
714 the writing process”) satisfies this requirement for the manuscript text itself.
715 We additionally extend disclosure, on our own initiative, to the experimental
716 adjudication pipeline described in this appendix, on the view that trans-
717 parency about automation matters most exactly where it could affect the
718 reported scientific conclusions, not only the prose describing them. The hu-
719 man adjudication gates of [Appendix C.3](#) are the mechanism by which we
720 take the responsibility the policy assigns to authors: every threshold, every
721 ruling on a caught defect, and every interpretive claim in this paper was
722 fixed or decided by a human author, with the automation’s role confined to
723 mechanical computation subject to audit and override.

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